

# When Self-Modification Becomes Memory

## An Audited Comparison of Three Self-Improving Agent Substrates

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### Abstract

Language-model agents can edit their own prompts, memories, tools, and code, but self-improvement bundles five separable claims: retention, expression, behavioral mediation, task improvement, and recursive improvement. We audit three persistent self-modification substrates: an arbitrary workspace, a success-gated procedure store, and a governed executable Pack. They share one outer agent, a 28-task development curriculum, cold-restart evaluation on 19 held-out tasks, matched ablation, sham, and cold controls, and immutable receipts. None of the three recorded development lineages met the preregistered uplift criterion. The shared interface expressed retained structure almost entirely as text. Within-substrate per-task item coverage ranged from 12% to 100%, and the actor-visible action schema exposed no retained invocation handle. Six no-context arms issued byte-identical initial requests per task yet differed on 7 of 19 tasks. Descriptive family-level mean gaps were 2/10 tasks (20.0 points) on SWE, 1/6 task (16.7 points) on Terminal, and 21.3 points on continuous-score ActiveGraph. An approximately byte-size-matched sham scored below matched ablation, so evolved-minus-sham contrasts could not distinguish sham interference from ordinary outcome spread. Mechanism audits identified vocabulary-driven failure-memory stickiness, append-only capacity saturation, and common-template reflection-proposal homogenization. Two independent outcome-blind model-coding passes agreed that all 84 proposals shared one nonexecutable textual genre; interpretive labels showed lower agreement and remain exploratory. We contribute an expression profile, an equivalent duplicates protocol, and a control ladder. The interface and acceptance mechanism are part of the self-improving system and must be measured.

## 1 Introduction

Language-model agents now edit their own prompts, memories, tools, and source code, and the reported results are striking. STOP [1] recursively improves its own improver. SICA [2] raises its score on a SWE-bench Verified subset from 17% to 53%. The Darwin Gödel Machine [3] more than doubles SWE-bench performance across a branching archive of self-modified descendants, and the Huxley-Gödel Machine [4] scores lineages by the productivity of their descendants. These systems make self-modification

easy to observe. Self-improvement is a different empirical claim. A visible edit establishes that the system changed itself. It does not establish that the change was expressed at evaluation time, altered behavior, caused better externally graded outcomes, or improved the system’s capacity to produce useful future changes.

We separate the statement that an agent improved itself into five constructs: (1) **retention**, meaning persistent state changed; (2) **expression**, meaning retained state reached the evaluation-time policy; (3) **behavioral mediation**, meaning the policy’s trajectory or resource use changed; (4) **task improvement**, meaning externally graded held-out performance rose beyond matched controls and stochastic spread; and (5) **recursive improvement**. Each construct requires different evidence. A score delta addresses task improvement. The other links require state, exposure, trajectory, and lineage evidence.

We study those links in a heavily instrumented comparison of three persistent self-modification substrates chosen for the breadth of their native update surfaces: an arbitrary rewritable workspace, a success-gated procedure store, and a governed, hash-pinned executable ActiveGraph Pack. Each traversed the same ordered 28-task development curriculum and then faced 19 disjoint held-out tasks across three families after a cold restart. The design used external graders, four arms, immutable receipts, hash-bound adjudication, and a preregistered success criterion. It also had one development lineage per architecture, one model, and small task families. The orchestration and local ActiveGraph family were self-built; SWE and Terminal used external official graders, and every claim is scoped to the recorded harness. The preregistered result is null: none of the three recorded lineages met the criterion. This paper explains why that null is informative and why several favorable and unfavorable results inside our data dissolve under audit.

The audit measures mechanisms that recur across agent systems. First, budgeted context is a common route from memory to later action. Under that route, our three substrate-interface configurations delivered between 12% and 100% of their declared retained items per task, exposed no retained invocation handle in the actor-visible action schema, and expressed native structural change to the acting policy almost entirely as text. These percentages are within-substrate coverage measures rather than comparable quantities of useful information. Second, single-run task estimates can conceal substantial outcome variability. Our six no-context labels issued byte-identical first requests per task, seed included, and still differed on 7

of 19 tasks. Their family-level means spanned 2/10 tasks (20.0 points) on SWE and 1/6 task (16.7 points) on Terminal; the continuous-score ActiveGraph span was 21.3 points. Third, placebo context can change the computation it is meant to control. On SWE, the Workspace sham solved 6/10 tasks and the matched ablation solved 8/10, creating a 20-point evolved-minus-sham contrast; the six equivalent no-context labels on the same ten tasks already ranged from 6/10 to 8/10. Fourth, lexical retrieval and append-only retention created observable selection and capacity effects. A top-three retriever over-selected broad-vocabulary failure lessons, and the Pack eventually blocked three consecutive updates at its policy limit.

The shared reflection layer created a proposal-form bottleneck before architecture-specific acceptance and native mutation. Two independent outcome-blind model-coding passes over all 84 update proposals assigned every proposal to the same representational genre: natural-language, append-only, nonexecutable, operational guidance with no specified activation. Interpretive fields were less stable across coders and remain exploratory pending independent human coding. The replicated form finding concerns the proposal layer. Native Workspace edits and Pack source changes remained structurally real and heterogeneous.

The receipts cut in both directions. An apparent transfer success involved an evolved SymPy pass matched by a no-context run that produced essentially the same fix, while the retained SymPy lesson concerned a different mechanism. An apparent retrieval failure scored exactly at the minimum of six equivalent no-context runs. The one evolved result outside its task’s exact equivalent range was a regression in which retained context mediated a larger invalid intervention and the ablation produced the smaller correct fix. A case audit that can reject favorable and unfavorable causal stories symmetrically is a minimum requirement for interpreting individual trajectories.

Prior systems provide strong evidence for the forms of optimization they measure. Table 6 records which studies report held-out transfer, mechanism ablations, sham retained state, independent evolution runs, byte-identical initial-request duplicates, immutable receipts, and descendant productivity. We do not reinterpret their reported gains through our experiment. Under our controls, our apparent gains dissolved. It is currently unknown how many published self-improvement effects would survive them.

This paper contributes six reusable instruments and observations:

1. a five-construct decomposition separating retention, expression, behavioral mediation, task improvement, and recursive improvement;
2. an **expression profile** measuring storage reach, eligible-payload reach, semantic-unit reach, selection adaptivity, and execution reach;
3. an **equivalent duplicates** protocol that creates task-level empirical dispersion references from executions with byte-identical initial requests;
4. a **control ladder** organizing eight control designs by

- the nuisance they address and the claims they license;
5. mechanism checks for **failure-memory stickiness**, capacity saturation, and common-template reflection-proposal homogenization; and
6. a fully receipted null with traces, graders, requests, artifacts, and an adjudication ledger.

The scope is a within-study diagnostic demonstration. It does not supply a population estimate, a universal architecture ranking, or a universal auditability-expressivity law. The expression bottleneck was created by the implemented common interface, and that fact is central to the result. The reflection template determined what could be proposed, the acceptance gate determined what survived, the retriever and adapter determined what was expressed, and the execution stack determined what a score could mean. The interface and the acceptance mechanism are part of the self-improving system.

## 2 Measurement framework

Let  $R_i^t$  denote retained state for architecture  $i$  after development generation  $t$ . Let  $A_i(R_i^t, x)$  be the evaluation adapter for task  $x$ ,  $O_i^t(x)$  the observation and action surface emitted by the adapter,  $\pi(O_i^t(x), x)$  the induced acting policy,  $Y_i^t(x)$  the external task score, and  $M_i^t$  the productivity of the process that creates useful descendants. The five constructs correspond to different comparisons:

1. Retention:  $R_i^{t+1} \neq R_i^t$ .
2. Expression:  $O_i^{t+1}(x) \neq O_i^t(x)$ .
3. Behavioral mediation: the induced policy trajectory changes.
4. Task improvement: expected external score increases beyond matched controls and relevant outcome spread.
5. Recursive improvement: the productivity  $M_i^t$  of generating useful descendants increases.

This decomposition prevents evidence from moving silently between levels. A new file establishes retention. A serialized lesson establishes possible expression. A changed trace establishes behavioral mediation. An external score contrast supports task improvement. Recursive improvement is the construct; descendant productivity is its operationalization. It requires evidence that earlier changes improved later update production.

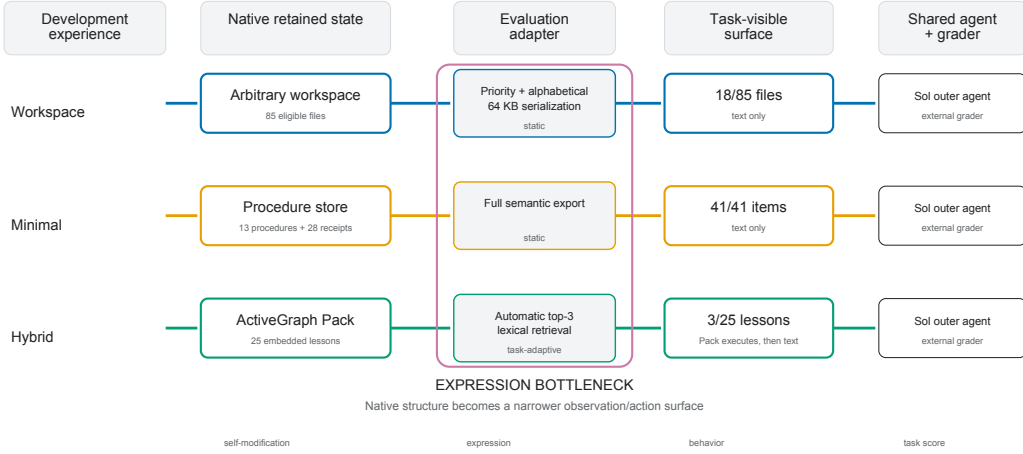
### 2.1 Expression profile

A single byte ratio cannot compare heterogeneous stores. Minimal’s complete SQLite state is mostly provenance, while its semantic export is small. Workspace bytes mix source, tests, instructions, and records. Hybrid executes a Pack query before the actor, so direct byte delivery omits runtime activation. We therefore report five structured axes:

1. **storage reach**: unique bytes copied into the initial actor-visible context divided by bytes in a declared canonical serialization of the complete frozen retained-state snapshot;

Figure 1. The expression bottleneck

Persistent change is filtered before it can affect action; each arrow is a distinct empirical claim.



**Figure 1: The expression bottleneck.** The study separates retention, expression, behavioral mediation, task improvement, and recursive improvement. Retention is necessary but does not identify later score improvement or recursive self-improvement. The bold bottleneck is the implemented adapter boundary. It does not imply that all auditable interfaces reduce expressivity.

2. **eligible-payload reach**: those copied bytes divided by the canonical retained byte spans admitted by the adapter’s frozen eligibility rule;
3. **semantic-unit reach**: declared substrate-native retained items surfaced per task, plus their union across the probe, divided by the frozen item inventory;
4. **selection adaptivity**: whether selection is explicitly conditioned on task input, accompanied by the per-task-to-probe-union coverage expansion; and
5. **execution reach**: whether retained code runs automatically and how many retained invocation handles appear in the actor-visible action schema.

Semantic units are files for Workspace, typed procedures, capabilities, or receipts for Minimal, and lessons for Hybrid. The unitization rule and item types are frozen before evaluation. These ratios measure coverage within one substrate and must not be used to compare quantities of useful information across substrates. The coverage-expansion gap measures selection turnover; it does not by itself show that selection is relevant or beneficial. Interfaces without byte-copy semantics should report the byte axes as not applicable and declare a proxy rather than force a ratio.

The expression profile is diagnostic rather than ordinal. Greater reach can increase useful transfer, interference, cost, or all three.

## 2.2 Control ladder

The control ladder organizes controls by the nuisance they address and the inference they license. Rungs 3 through 6 are alternative payload controls, not a monotonic sequence in which every higher number dominates every lower number.

The present study implements rungs 1, 2, 3, and 7. The

other rows are operational specifications for follow-up designs rather than empirical controls claimed here.

## 3 Study design

All three approaches used the same `gpt-5.6-sol` outer coding agent, brokered tool protocol, task ceilings, grader boundary, and common reflection step. They differed in retained state:

- **Workspace v1.2** could rewrite and retain an arbitrary workspace tree.
- **Minimal v2** stored every scalar receipt and promoted only successful experiences as procedures or deterministic capabilities.
- **Hybrid Packs** used a model-authored, tested, hash-pinned ActiveGraph Pack to store and retrieve lessons.

The native mutation units differ in breadth. The score comparison concerns their implemented common interface and does not order their native ceilings.

Each architecture traversed the same 28-task development curriculum: 20 SWE-bench Verified issues, six Terminal-Bench 2 tasks, and two local Ouro-ActiveGraph-50 systems. Evaluation used 19 disjoint held-out tasks: ten SWE-bench Verified issues, six Terminal-Bench 2 tasks, and three Ouro-ActiveGraph-50 systems. SWE and Terminal used official binary verifiers. The ActiveGraph family used normalized scores from 50 sealed checks.

For each architecture and held-out task, the study ran four arms:

- **evolved**, with retained development state enabled;
- **cold ablation**, with the same lineage present and learned state hidden;
- **sham**, with a seeded opaque artifact intended to approximate the evolved context size; and
- **cold**, with no development state.

**Control ladder.** Payload controls 3–6 are alternatives rather than a monotonic sequence.

| Control                       | Intervention and held-fixed variables                                                                               | Licensed inference                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1. Cold floor                 | Same model, task, harness, and budget; no lineage                                                                   | Model-and-harness floor                                              |
| 2. State-hidden ablation      | Same developed lineage and evaluation stack; learned state hidden                                                   | Effect of enabling the implemented retained-state channel            |
| 3. Opaque byte-size sham      | Seeded opaque text; report byte ratio and token load                                                                | Sensitivity to opaque payload of recorded size                       |
| 4. Token-matched neutral sham | Match tokenizer, position, and count with a preregistered corpus and blinded relevance screen                       | Payload effect after equalizing first-call token load                |
| 5. Semantic sham              | Plausible units from disjoint task families under blinded relevance screening                                       | Sensitivity to screened-unrelated guidance                           |
| 6. Structure-preserving sham  | Preserve count, schema, and token budget while permuting content or metadata                                        | Sensitivity to retained structure apart from its original mapping    |
| 7. Equivalent duplicates      | Byte-identical initial request, model, seed field, task image, harness, and budgets; later trajectories may diverge | Observed within-task execution dispersion                            |
| 8. Native-interface ablation  | Hold state, task, model, and budgets fixed while toggling native execution                                          | Effect of the native execution path with recorded downstream changes |

The preregistered primary estimand was paired **evolved - cold ablation**. Strong recursive uplift required benefit over ablation and sham, a nominal 95% task-resampling interval above zero in at least two families including SWE or Terminal, valid cold reload, and no material development regression. This was an operational decision rule for the recorded probe rather than a claim of population-level interval coverage.

The information boundary excluded hidden tests, expected outputs, private cases, grader logs, and oracle artifacts from lineage state. Each accepted generation produced an immutable directory whose manifest bound architecture, parent run, development experience, artifact size, and hash. Evaluation occurred after cold reload. The frozen repository commit was 559843894578141dfbd39b7bae2246db76e57bde.

The study completed 84 development attempts and 228 held-out attempts. Failed attempts remained in the denominator. Six SWE runs reached the frozen 3,600-second task limit. Six Terminal agents reached the 1,800-second limit, and completed verifiers returned zero. One Active-Graph grader contradicted a receipt-bound parsable submission; one sealed grader-only retry of the hash-identical artifact recovered 24/50. No model attempt was rerun and no raw row was rewritten. The adjudication index binds the raw report, datasets, changed analytical row, and evidence by SHA-256.

Scores remain separate by family. Nominal 95% descriptive task-resampling intervals use 10,000 deterministic percentile-bootstrap draws. Each draw resamples the paired per-task arm differences jointly with replacement. With one lineage and ten, six, and three held-out tasks, no frequentist coverage is asserted. The three-task Active-Graph interval is especially discrete and unstable.

## 4 Preregistered outcome

None of the three recorded lineages met the preregistered uplift criterion. Table 1 reports the binary families as fractions and percentages. For context, the six equivalent no-context labels ranged from 6/10 to 8/10 on SWE and from 4/6 to 5/6 on Terminal despite byte-identical initial

requests within each task.

The table establishes persistent evaluation after development and the absence of preregistered uplift. It does not explain whether retained state was available, whether it changed behavior, or whether visible contrasts exceeded outcome variability. Sections 5 through 7 audit those links.

## 5 Expression bottleneck

All three architectures created cold-loadable state:

These receipts establish retention. The expression profile shows what could reach later behavior:

Minimal supplies the decisive counterexample to storage reach as a sufficient metric. Its database and provenance artifacts were large, giving 1.35% storage reach, while the adapter exported all declared semantic units. Those 41 units comprised 13 procedures and 28 receipts. None was a deterministic capability.

Workspace exposed 17 complete files and part of an eighteenth. All post-priority files came from the alphabetically early `_development/` directory, and the actor could not invoke retained tools. Minimal delivered the same complete export to every task. Hybrid alone selected different learned units by task: three of 25 per task and 17 of 25 across the probe. Its Pack executed a retrieval before each task, while the actor received only the returned text and could not call the Pack interactively.

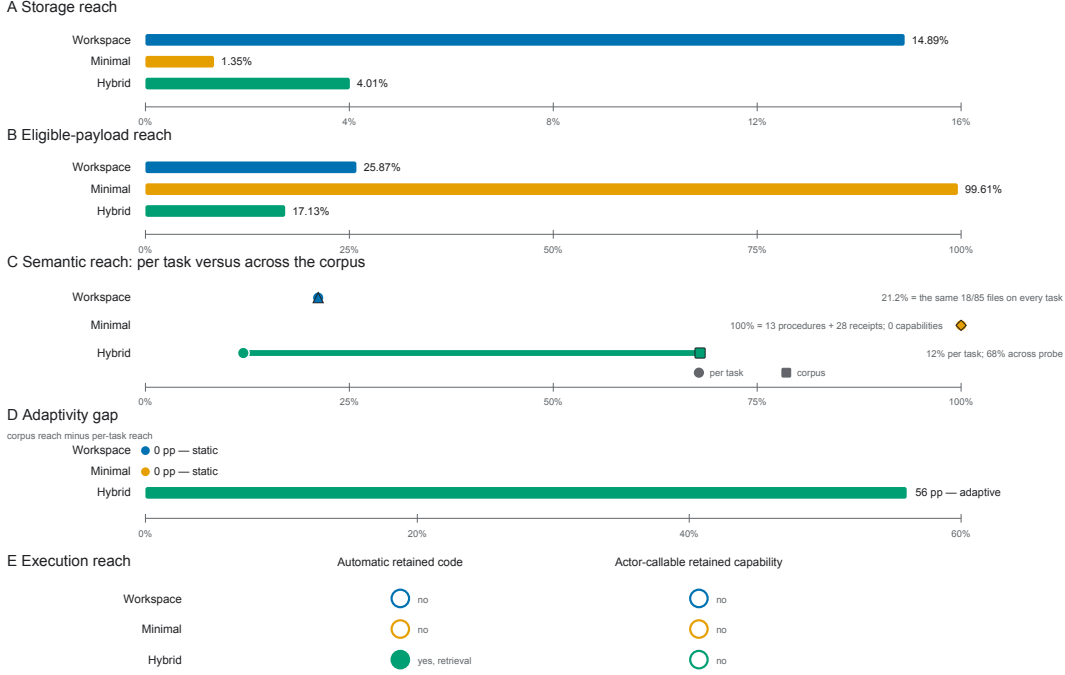
The common adapter therefore narrowed each native substrate in a different way. Workspace was truncated and static. Minimal was semantically complete and static. Hybrid used task-conditioned selection with 56 points of probe-union coverage expansion. No actor-visible action schema contained a retained invocation handle. Minimal had no retained executable unit, while Workspace and Hybrid retained executable material that the actor could not invoke on demand. This supports an expression bottleneck in the implemented interface. A general auditability-expressivity tradeoff remains a hypothesis for a future interface intervention.

**Table 1:** Preregistered held-out outcomes. Intervals are nominal descriptive task-resampling intervals.

| Family          | Architecture | Evolved | Ablation | Evolved-ablation [interval] | Sham   | Evolved-sham |
|-----------------|--------------|---------|----------|-----------------------------|--------|--------------|
| SWE (10)        | Workspace    | 8/10    | 8/10     | 0/10 [0, 0]                 | 6/10   | +2/10        |
|                 | Minimal      | 7/10    | 6/10     | +1/10 [-20, +40] pp         | 7/10   | 0/10         |
|                 | Hybrid       | 8/10    | 7/10     | +1/10 [0, +30] pp           | 8/10   | 0/10         |
| Terminal (6)    | Workspace    | 5/6     | 4/6      | +1/6 [0, +50] pp            | 3/6    | +2/6         |
|                 | Minimal      | 3/6     | 5/6      | -2/6 [-66.7, 0] pp          | 4/6    | -1/6         |
|                 | Hybrid       | 4/6     | 4/6      | 0/6 [0, 0]                  | 4/6    | 0/6          |
| ActiveGraph (3) | Workspace    | 82.7%   | 100.0%   | -17.3 [-52, 0] pp           | 96.0%  | -13.3 pp     |
|                 | Minimal      | 96.7%   | 88.0%    | +8.7 [-10, +36] pp          | 100.0% | -3.3 pp      |
|                 | Hybrid       | 82.7%   | 92.7%    | -10.0 [-40, +10] pp         | 82.7%  | 0.0 pp       |

**Figure 2.** Expression profiles are multidimensional

Minimal’s 1.35% storage reach coexists with 100% semantic reach; no single byte ratio is sufficient.



**Figure 2: Expression is a profile.** Five axes are reported: storage reach, eligible-payload reach, semantic coverage with per-task and probe-union submeasures, selection adaptivity, and execution reach with automatic and actor-visible submeasures. Percentages are within-substrate coverage measures and do not compare quantities of useful information. Minimal’s 41 declared units are 13 procedures and 28 receipts; none is executable. Hybrid conditions top-three selection on task text, exposing three of 25 lessons per task and 17 across the probe. Its Pack runs automatically in the adapter and supplies no on-demand actor-visible invocation handle.

**Table 2:** Retained state after 28 development tasks.

| Architecture | Accepted after |      | Final retained product                     |
|--------------|----------------|------|--------------------------------------------|
|              | Pass           | Fail |                                            |
| Workspace    | 13             | 12   | 429,731-byte workspace                     |
| Minimal      | 13             | 0    | 13 procedures, 0 capabilities, 28 receipts |
| Hybrid       | 15             | 10   | 25 lessons; 62,506-byte Pack               |

## 6 Equivalent duplicates and control interference

The six no-context labels were Workspace cold, Workspace ablation, Minimal cold, Minimal ablation, Hybrid cold, and Hybrid ablation. For each task, their first

model requests were byte-identical, including seed. Their outcomes varied on 7 of 19 tasks. The remaining 12 tasks were stable: ten fixed passes and two fixed failures.

The one-way random-effects ICC treats tasks as targets and the six named no-context labels as exchangeable single replicate executions on the observed score scale. It is a descriptive repeatability coefficient, and the three-target ActiveGraph value is especially unstable. The design-size diagnostic is  $(1.96 + 0.842)\widehat{SE}$ , using a two-sided nominal  $\alpha = 0.05$ , 80% power, and the within-task variance across the six equivalent labels. It is a continuous normal approximation for a paired mean contrast. On the binary families it does not represent an attainable observed score increment.

All discriminative variation in the no-context probe was

**Table 3:** Expression profile. Coverage ratios are within-substrate and do not compare useful information.

| Architecture | Storage   | Eligible   | Semantic: task; union   | Selection; expansion             | Execution: automatic; handles   |
|--------------|-----------|------------|-------------------------|----------------------------------|---------------------------------|
| Workspace    | 14.9%     | 25.9%      | 18/85 (21.2%); 18/85    | static; 0 pp                     | none; 0, retained tools hidden  |
| Minimal      | 1.35%     | 99.6%      | 41/41 (100%); 41/41     | static; 0 pp                     | none; N/A, 0 executable units   |
| Hybrid       | 4.0% mean | 17.1% mean | 3/25 (12%); 17/25 (68%) | task-conditioned; 56 pp turnover | automatic Pack query; 0 handles |

**Table 4:** Equivalent duplicate-execution dispersion and design-sizing diagnostics.

| Family      | Tasks | Variable | ICC(1,1)       | Maximum pair gap | Nominal 80% MDE |
|-------------|-------|----------|----------------|------------------|-----------------|
| SWE         | 10    | 3        | 0.72           | 2/10 (20.0 pp)   | 30.7 pp         |
| Terminal    | 6     | 2        | 0.66           | 1/6 (16.7 pp)    | 45.1 pp         |
| ActiveGraph | 3     | 2        | 0.31, unstable | 21.3 pp          | 28.7 pp         |

concentrated in seven tasks that changed outcome under equivalent requests. Stable tasks still provide one-sided opportunities: fixed passes can expose regressions and fixed failures can expose gains. Minimal’s evolved failure on `scikit-learn__scikit-learn-13124`, against six equivalent passes, was the only one of 57 evolved architecture-task results outside an exact task-specific equivalent range.

The maximum of the 15 absolute pairwise label-mean gaps is reported for each family. The 15 values are dependent functions of six labels, and the labels are not randomized evolution replications. The maximum is a descriptive dispersion reference rather than a significance threshold.

The sham comparison exposes a second control problem. Workspace SWE evolved minus sham was +2/10 tasks (+20.0 points) because the evolved and ablation arms each solved 8/10 tasks while the opaque sham solved 6/10; equivalent no-context labels on the same tasks also ranged from 6/10 to 8/10. On Terminal, Workspace evolved minus sham was +2/6 tasks (+33.3 points) because evolved solved 5/6, ablation solved 4/6, and sham solved 3/6; equivalent labels on those tasks ranged from 4/6 to 5/6. The SWE sham family score fell within the equivalent-label range, at 6/10 against 6/10 to 8/10. The Terminal sham fell below all six equivalent family means, at 3/6 against 4/6 to 5/6. With six dependent draws, a seventh observation below the observed minimum is consistent with interference and does not establish it. The sham consumed approximately 1.88 to 2.43 times as many incremental first-call tokens per byte as evolved context, depending on architecture. Its context bytes exactly matched evolved context for Workspace and Minimal. For Hybrid, the 6,093-byte sham was 56% to 58% of the mean evolved context across families. Byte matching therefore controlled artifact size while leaving tokenization, semantic load, and interference uncontrolled for Workspace and Minimal; Hybrid also retained a substantial byte-size mismatch.

## 7 Mechanism audits

### 7.1 Failure-memory stickiness

Hybrid retained 25 lessons, including 15 from passing development attempts and ten from failed attempts. Across 19 held-out tasks, the top-three retriever filled 57 slots. Failure-derived lessons occupied 37/57 slots (64.9%) despite forming 10/25 of the corpus (40%). Two failed Pylint lessons occupied 22/57 slots (38.6%). Only 17/25 lessons were ever retrieved, and 5/57 slots shared a repository prefix with the held-out task.

The implemented ranker counted set overlap between task words and an index containing suite, task identifier, title, scope, and trigger terms. Failure-derived lessons had broader indexed vocabularies. Indexed vocabulary correlated with retrieval count within passing lessons ( $\rho = 0.743, n = 15$ ) and failed lessons ( $\rho = 0.763, n = 10$ ). In the descriptive regression `retrievals ~ indexed vocabulary + failure flag`, vocabulary had a coefficient of 0.194 retrievals per additional term and the adjusted failure coefficient was -0.093 retrievals. A 50,000-draw descriptive within-status label-permutation diagnostic shuffled vocabulary values within pass/fail strata, re-fit the vocabulary coefficient, and counted absolute coefficients at least as large as observed with a plus-one Monte Carlo correction. It gave  $p_{MC} \approx 0.0045$ . This conditional reference requires exchangeability of vocabulary labels within status; the 25 lessons form one fixed corpus and compete for 57 retrieval slots, so the value is neither causal nor population inference.

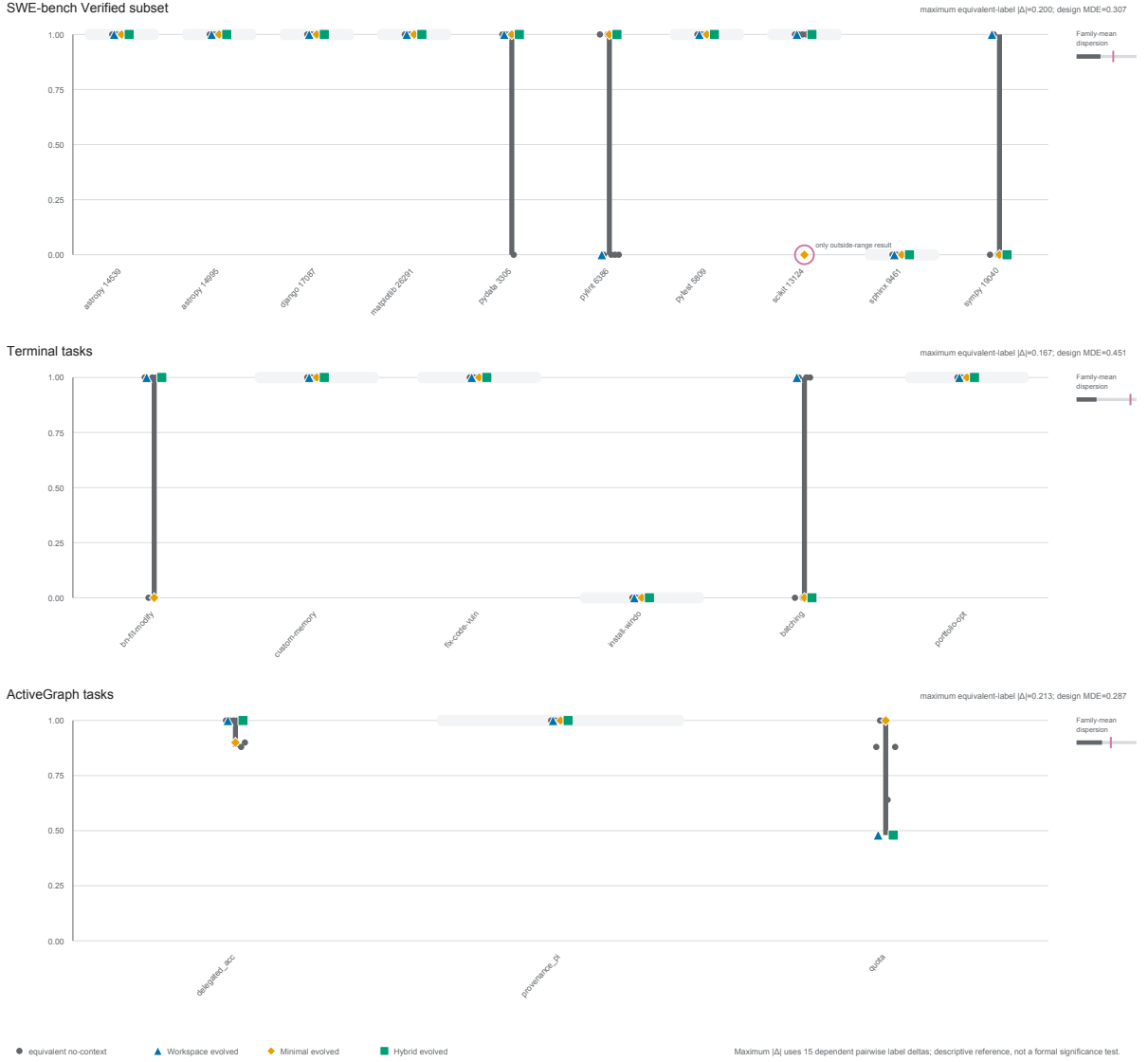
Offline re-ranking the same prompts changed the selection pattern without new model calls. Failure slots fell from 37/57 under raw overlap to 33/57 under Jaccard, 27/57 under binary cosine, and 28/57 under BM25. BM25 surfaced 22/25 lessons and reduced the top-two share from 22/57 to 10/57. These diagnostics identify vocabulary coverage as a mechanism in the implemented ranker. They do not show that alternative retrieval improves downstream task scores.

### 7.2 Capacity saturation

Hybrid Pack source grew from 7,053 to 62,506 bytes. Generation 25 first exceeded the 64,000-byte policy and succeeded after one repair. Every candidate in generations

Figure 3. Evolved outcomes versus task-specific duplicate-execution clouds

Six byte-identical no-context requests define each gray cloud; Minimal/scikit-learn is the only evolved result outside a cloud.



**Figure 3: Task-level initial-request duplicate spread.** For each of 19 tasks, gray points show six no-context outcomes generated from byte-identical initial model requests; colored marks show the three evolved outcomes. Later trajectory inputs may diverge. Seven tasks varied under the duplicate executions and 12 were stable. Minimal/scikit-learn is the only evolved architecture-task outcome outside the observed duplicate-execution range. The bar is the maximum of 15 dependent absolute pairwise label-mean gaps and is descriptive rather than a formal significance threshold.

26 through 28 remained over the cap after four author or repair requests; the closest final candidate was 65,928 bytes. Mean authoring requests rose from 1.0 over the first five generations to 3.2 over the last five, and mean authoring cost rose from \$0.249 to \$2.114.

The policy rejection and retained-state plateau directly establish saturation under this append-oriented design. Task identity and difficulty are confounded with generation order, so the cost trend is descriptive. The result does not establish a general limitation of ActiveGraph or executable memory.

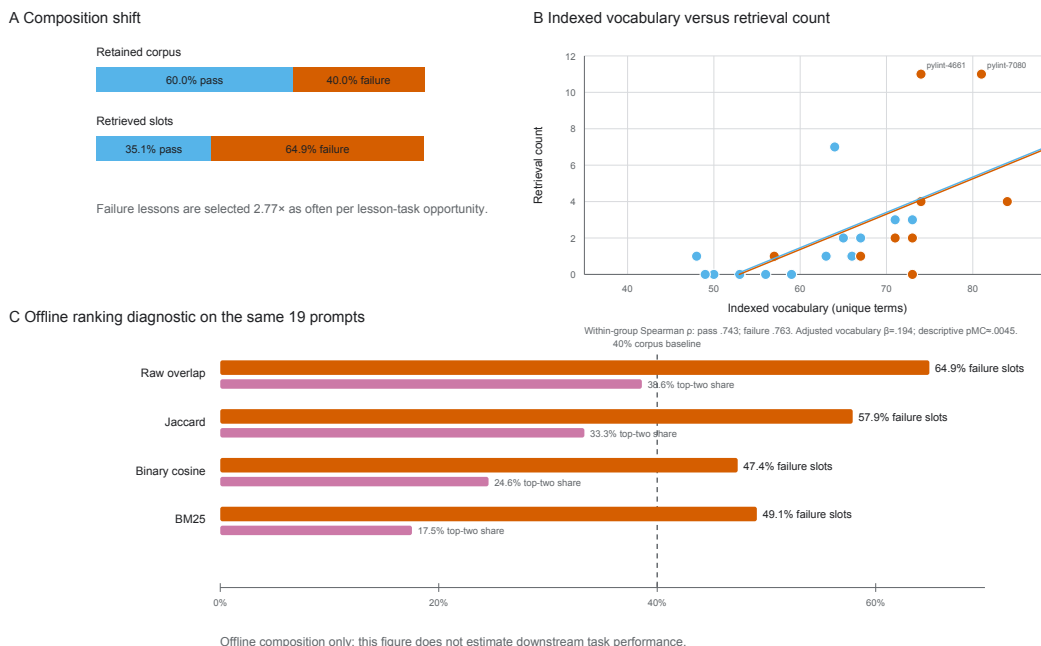
### 7.3 Update-proposal taxonomy

A frozen codebook was applied to all 84 generation-level reflection proposals. Each coder saw opaque shuffled excerpts without architecture, task, sequence, score, acceptance, cost, or later retrieval metadata. Sol and Terra completed independent high-reasoning coding passes. Original labels were hashed before comparison and no disagreement was adjudicated.

Both coders labeled 84/84 proposals as natural-language, append-only, nonexecutable, operational guidance with no specified activation. They agreed on failure specificity for 83/84 proposals and validation strategy for

**Figure 4. Failure-memory stickiness is a vocabulary-coverage mechanism**

Failures are 40% of memory but 64.9% of retrievals; vocabulary predicts selection within both outcome groups.



**Figure 4: Failure-memory stickiness and lexical coverage.** Hybrid’s retained corpus was 40% failure-derived, while failures occupied 37/57 retrieval slots (64.9%). Retrieval count increased with indexed vocabulary within both success and failure groups. The adjusted regression assigns nearly all linear signal to vocabulary breadth rather than failure status. This identifies a mechanism in the implemented ranker and does not establish an effect on held-out score.

82/84. These replicated labels support a stable description of proposal form.

Interpretive fields were less stable: agreement was 59/84 for abstraction, 59/84 for evidence grounding, 63/84 for transfer scope, and 47/84 for secondary target. Both coders assigned the same grounding totals to success-derived proposals, 39/42 direct-validated and 3/42 inferred. For failure-derived proposals, Sol assigned 7/42 direct-validated, 5/42 direct-unvalidated, and 30/42 inferred; Terra assigned 21/42, 6/42, and 15/42. The direction of the success-failure contrast replicated and its magnitude did not.

The shared reflector therefore produced a homogeneous representational genre. Architecture-specific gates selected different retained mixtures: Minimal accepted 13 success-derived proposals and no failure-derived proposals; Workspace accepted 25 proposals including 12 after failures; Hybrid accepted 25 including ten after failures. Human inter-rater reliability remains pending, and interpretive distributions remain exploratory. The analysis describes common update proposals rather than every native Workspace edit or Pack source change.

## 8 Audited cases

Case selection followed a frozen rule that required a complete chain:

retained update -> expression -> plausible mechanism -> later behavior

The rule also required comparison with the exact task-specific range among equivalent duplicates.

**Apparent success.** Workspace passed a held-out SymPy task. Its retained SymPy lesson concerned a different mechanism, and a no-context run generated essentially the same fix. Equivalent outcomes on the task ranged from failure to pass. The receipts establish a pass and reject the learned-transfer story.

**Apparent failure.** Hybrid scored 0.48 on quota scheduler after retrieving mechanistically skewed lessons. Six equivalent no-context runs on the same task ranged from 0.48 to 1.00. The evolved score equaled the observed minimum. The receipts establish retrieval skew and reject attribution of this outcome to that skew.

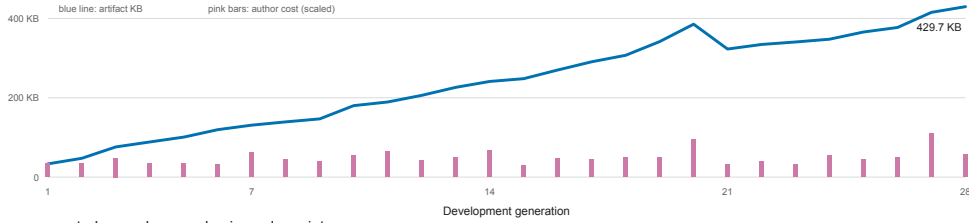
**Outside-range regression.** Minimal failed a scikit-learn task that all six equivalent no-context runs passed. The retained-context trajectory changed an algorithm and rewrote an established expected-output test. The ablation made a smaller fix and passed the restored official test. This establishes harmful behavioral mediation. The trace does not identify one retained sentence as the cause.

**Capacity ceiling.** Hybrid generation 25 succeeded after a repair near the policy cap. Generations 26 through 28 produced four over-cap candidates each. This case establishes a direct mechanism from append-only accumu-

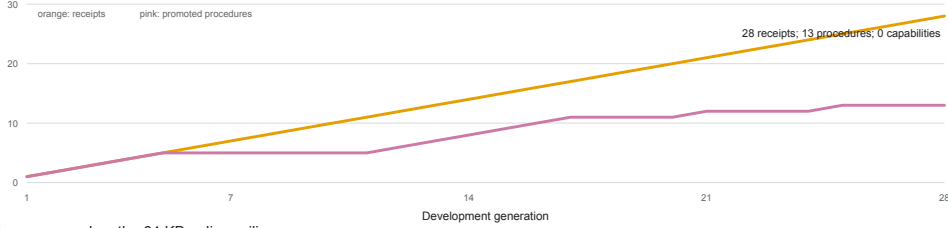
Figure 5. Accumulation, selective promotion, and saturation

All substrates retained state; only Hybrid encountered an explicit capacity ceiling.

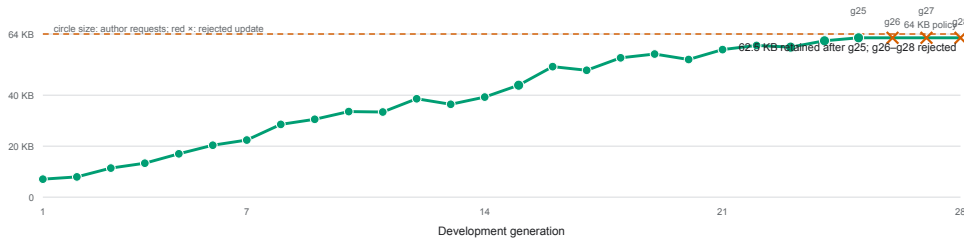
A Workspace: artifact growth and authoring cost



B Minimal: success-gated procedures and universal receipts



C Hybrid: Pack source reaches the 64 KB policy ceiling



**Figure 5: Accumulation, rising effort, and saturation.** Panel A shows Workspace artifact growth and authoring cost; panel B shows Minimal’s universal receipts and success-gated procedures; panel C shows Hybrid Pack growth and rejected updates at the 64,000-byte source policy. Hybrid generation 25 succeeded after repair and generations 26 through 28 produced no within-cap candidate. The cost trend is descriptive because generation order is confounded with task identity and difficulty.

lation to failed retention.

Together, the cases show why receipts must be permitted to invalidate stories in both directions. They also show why score, mechanism, and attribution should be reported separately.

## 9 Related work

Self-improving agent systems measure different portions of the five-construct chain. STOP reports repeated recursive improver optimization and held-out task utilities. Gödel Agent [5] evaluates runtime self-referential modification with tool and policy ablations. SICA optimizes a coding-agent source tree through a reported primary lineage. Darwin Gödel Machine uses a branching archive and reports transfer across benchmarks, models, and languages. Huxley-Gödel Machine explicitly measures descendant productivity through clade-metaproductivity.

Our study complements these capability results by instrumenting retained-state attribution. Table 6 separates unevolved baselines, mechanism interventions, independent complete evolution runs, versioned provenance, immutable exposure receipts, and recursive-improvement operationalization. No design reports every listed feature.

This study lacks independent evolution-lineage replication and does not identify descendant productivity. It adds sham state, initial-request duplicates, and immutable exposure receipts to a persistent post-curriculum comparison.

Reflexion [6] shows that episodic linguistic feedback can alter later trials; AI Agents That Matter [7] motivates cost, holdout, and attribution controls. Stochasticity in Agentic Evaluations [8] motivates consistency analysis; PACE [9] studies noisy acceptance. Useful Memories Become Faulty When Continuously Updated by LLMs [10] contrasts episodes with consolidation; DeltaMem [11] studies residual-tree memory. Together they motivate measurement of retained-state mechanisms.

**Table 6: Related-work evidence features.** “Yes” means that the cited primary paper reports the feature. “Partial” identifies a nearby but weaker design. “NR” means not reported and does not imply that the authors could not implement it. Cells were re-audited against the linked primary paper versions on 2026-07-23. The strict receipt column requires immutable binding among retained state, evaluation exposure, trajectory, and outcome.

**Table 6:** Related-work evidence features, audited against primary papers on 2026-07-23. NR means not reported.

| Panel A: substrate, transfer, and interventions |                                           |     |                                             |       |                                        |  |                                             |                   |                                   |
|-------------------------------------------------|-------------------------------------------|-----|---------------------------------------------|-------|----------------------------------------|--|---------------------------------------------|-------------------|-----------------------------------|
| System                                          | Retained surface                          |     | Held-out transfer                           |       | Initial baseline                       |  | Mechanism                                   | interven-<br>tion | Sham state                        |
| STOP [1]                                        | Python prover/scaffold                    | im- | Yes: independent LPN and five new utilities |       | Seed improver                          |  | Nonrecursive and alter-<br>native improvers |                   | NR                                |
| Gödel Agent [5]                                 | Runtime logic/actions                     |     | Partial: held-out splits                    |       | Initial policy and manual<br>baselines |  | Initial-tool ablations                      |                   | NR                                |
| SICA [2]                                        | Coding-agent source tree                  |     | NR: utility reused in pri-<br>mary run      |       | Initial agent                          |  | NR                                          |                   | NR                                |
| DGM [3]                                         | Coding-agent branching<br>archive         |     | Cross-benchmark,<br>model, language         |       | Initial agent and manual<br>baselines  |  | Without<br>improvement/open<br>exploration  | self-             | NR                                |
| HGM [4]                                         | Coding-agent search tree                  |     | Benchmark and model<br>transfer             |       | Shared initial agent                   |  | Matched SICA/DGM se-<br>lection             |                   | NR                                |
| This study                                      | Workspace; proce-<br>dures/receipts; Pack |     | 19 disjoint                                 | post- | Cold floor and initial har-<br>ness    |  | State-hidden ablation                       |                   | Opaque sham; mismatch<br>reported |

| Panel B: replication, provenance, and recursive-improvement evidence |                                                     |           |                      |                        |                                   |         |                                                                    |              |                                 |
|----------------------------------------------------------------------|-----------------------------------------------------|-----------|----------------------|------------------------|-----------------------------------|---------|--------------------------------------------------------------------|--------------|---------------------------------|
| System                                                               | Complete<br>runs                                    | evolution | Identical<br>request | initial-<br>duplicates | Versioned<br>edge/provenance      | lin-    | Immutable<br>sure/trajectory<br>ceipsts                            | expo-<br>re- | Descendant productiv-<br>ity    |
| STOP                                                                 | Five GPT-4; 25 smaller-<br>model in key experiments |           | NR                   |                        | NR                                |         | NR                                                                 |              | NR                              |
| Gödel Agent                                                          | Six cycles per task                                 |           | NR                   |                        | NR                                |         | NR                                                                 |              | NR                              |
| SICA                                                                 | NR: one primary lineage                             |           | NR                   |                        | Yes: archive of<br>agents/results |         | NR                                                                 |              | NR                              |
| DGM                                                                  | Three complete runs in<br>stability analysis        |           | NR                   |                        | Traceable archive tree            |         | NR                                                                 |              | NR                              |
| HGM                                                                  | NR for repeated complete<br>searches                |           | NR                   |                        | Retained search tree              |         | NR                                                                 |              | Yes: metaproductivity<br>clade- |
| This study                                                           | No: one lineage per sub-<br>strate                  |           | Six labels per task  |                        | Immutable<br>tions/hashe          | genera- | State, exposure, request,<br>trajectory, grader, adjud-<br>ication |              | No: not identified              |

## 10 Discussion

The preregistered null does not establish that persistent self-modification cannot improve agents. It establishes that three retained substrates, under one common interface and one lineage each, did not meet the specified uplift criterion. The mechanism audit explains why a score-only summary would discard the most informative evidence.

First, native update breadth and evaluation-time expression are separate design dimensions. Workspace retained arbitrary files, Minimal retained procedures and receipts, and Hybrid retained executable Pack source. The actor received truncated static text, complete static text, or task-conditioned text. It received no callable retained capability. Evaluation therefore compared native substrates after a consequential common transformation.

Second, proposal, acceptance, retrieval, and expression form a pipeline. The common reflector produced one representational genre across all 84 proposals. Acceptance gates created different epistemic mixtures. Workspace and Hybrid retained failure-derived proposals that Minimal excluded. Hybrid’s lexical retriever then concentrated exposure on broad-vocabulary failure lessons. Each mechanism belongs inside the system boundary because each changes what a later actor can use.

Third, controls can confound contrasts. The opaque sham matched evolved bytes for Workspace and Minimal but not Hybrid, and it differed in tokenization and semantic load. Equivalent duplicates varied despite identical first requests. A positive evolved-minus-sham contrast can therefore combine useful expression, possible sham interference, and ordinary outcome spread. The control ladder makes these alternatives explicit and suggests the next diagnostic comparison.

Fourth, append-only retention carries a measurable maintenance burden. Hybrid reached a hard cap; Workspace and Minimal accumulated without consolidation. This observation motivates consolidation, pruning, and replacement as first-class update operations. It does not establish which operation would improve held-out performance.

The instruments generalize to persistent agents whose retained-state inventory, eligibility rule, adapter emissions, and actor-visible actions can be frozen and inspected. Systems without byte-copy semantics should declare a computable proxy or mark the byte axes not applicable. Equivalent duplicates can be retained whenever multiple labels issue byte-identical initial requests under a common execution contract. The control ladder can be selected according to the intended causal claim. The update-proposal taxonomy can be applied to shared reflectors while native state changes are analyzed separately. Receipts can bind retained artifacts, exposures, trajectories, graders, and adjudication.

## 11 Limitations

The study completed one development lineage per architecture. The six equivalent labels expose outcome variability and do not replace independent evolution replications. All approaches shared one model and outer agent, which may dominate architecture differences. The held-out families contained ten, six, and three tasks. ActiveGraph uncertainty estimates are especially unstable.

The common interface constrained native expression and exposed no retained interactive capability. This limits architecture ranking and creates the expression bottleneck examined in the paper. A general auditability-expressivity relationship requires an intervention that varies the interface while holding retained state fixed.

The sham targeted byte matching rather than token or semantic matching, and Hybrid retained a substantial byte-size mismatch. It remains useful as an interference control and cannot identify equal-compute semantic effects. Duplicate-label pairwise gaps are dependent and post-hoc. The task-level ranges are descriptive.

The update-proposal taxonomy used two language-model coders. Exact agreement on representational form does not substitute for independent human coding. Interpretive labels showed lower exact agreement and remain exploratory. Architecture masking may be incomplete because proposal form can reveal substrate.

The retrieval analysis identifies lexical selection mechanics and does not establish causal harm to score. Offline re-ranking produced no new behavioral outcomes. Experiment B remains closed because the paper does not require a causal retrieval-harm claim. Consolidation at generation 25 is a separate Ouroboros development experiment.

## 12 Conclusion

All three systems retained persistent changes. Their evaluation interface expressed those changes mainly as text, their controls exhibited substantial outcome spread and left interference uncontrolled, and none met the preregistered uplift criterion. The resulting null separates five claims that are often compressed into one: retention, expression, behavioral mediation, task improvement, and recursive improvement.

The central lesson is operational. Treat reflection templates, acceptance gates, retrievers, adapters, execution paths, and controls as components of the self-improving system. Measure each link with receipts and matched interventions. The interface and acceptance mechanism determine what self-modification can become.

**Artifact availability.** The version-controlled release includes the paper, generated tables, frozen coding artifacts, figures, audit scripts, and a self-verifying manifest. Complete trace archive DOI pending its separate disclosure scan and deposit.

## References

- [1] Eric Zelikman, Eliana Lorch, Lester Mackey, and Adam Tauman Kalai. Self-taught optimizer (STOP): Recursively self-improving code generation, 2023. <https://arxiv.org/abs/2310.02304>.
- [2] Maxime Robeyns, Martin Szummer, and Laurence Aitchison. A self-improving coding agent, 2025. <https://arxiv.org/abs/2504.15228>.
- [3] Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin gödel machine: Open-ended evolution of self-improving agents, 2025. <https://arxiv.org/abs/2505.22954>.
- [4] Wenyi Wang, Piotr Piękos, Li Nanbo, Firas Laakom, Yimeng Chen, Mateusz Ostaszewski, Mingchen Zhuge, and Jürgen Schmidhuber. Huxley-gödel machine: Human-level coding agent development by an approximation of the optimal self-improving machine, 2025. <https://arxiv.org/abs/2510.21614>.
- [5] Xunjian Yin, Xinyi Wang, Liangming Pan, Xiaojun Wan, and William Yang Wang. Gödel agent: A self-referential agent framework for recursive self-improvement, 2024. <https://arxiv.org/abs/2410.04444>.
- [6] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning, 2023. <https://arxiv.org/abs/2303.11366>.
- [7] Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. AI agents that matter, 2024. <https://arxiv.org/abs/2407.01502>.
- [8] Zairah Mustahsan, Abel Lim, Megna Anand, Saahil Jain, and Bryan McCann. Stochasticity in agentic evaluations: Quantifying inconsistency with intraclass correlation, 2025. <https://arxiv.org/abs/2512.06710>.
- [9] Zayx Shawn. PACE: Anytime-valid acceptance tests for self-evolving agents, 2026. <https://arxiv.org/abs/2606.08106>.
- [10] Dylan Zhang, Yanshan Lin, Zhengkun Wu, Yihang Sun, Bingxuan Li, Dianqi Li, and Hao Peng. Useful memories become faulty when continuously updated by LLMs, 2026. <https://arxiv.org/abs/2605.12978>.
- [11] Haoran Tan, Zeyu Zhang, Zhicheng Cao, Rui Li, and Xu Chen. DELTAMEM: Incremental experience memory for LLM agents via residual trees, 2026. <https://arxiv.org/abs/2606.03083>.

## A Adjudication sensitivity

The raw run contained 215 immediately valid rows and 13 rows requiring policy adjudication. The final dataset retained all 228 attempts. Categories were six official SWE task timeouts, six Terminal agent-budget timeouts with completed zero verifiers, and one ActiveGraph snapshot contradiction resolved by a sealed grader-only retry of the hash-identical submission.

Eight of nine preregistered **evolved - cold ablation** contrasts are identical when raw invalid rows are scored zero and when the frozen adjudication decisions are applied. The exception is Hybrid ActiveGraph: the contrast moves from -26.0 points under raw-invalid-as-zero to -10.0 points after the hash-identical quota-scheduler regrade. It remains negative and the preregistered conclusion is unchanged. A raw complete-case analysis gives +7.3 points for this cell because it drops the evolved invalid row. Complete-case contrasts are reported only as sensitivity analyses because dropping bounded failures changes the estimand and can favor arms with more invalid outcomes.

The study’s single adjudication judgment moved a score in the evolved arm’s favor, and the preregistered conclusion remained null.

Machine-readable table: `data/generated/adjudication-sensitivity.csv`

## B Timeout taxonomy

Timeouts remain outcomes under the bounded-agent estimand. The taxonomy separates agent budget exhaustion, official evaluator limits, and infrastructure contradiction.

## C Success-conditional efficiency

For the binary SWE and Terminal families, the companion table reports mean cost and wall time among successful attempts for every architecture and arm: `data/generated/success-conditional-efficiency.csv`.

These quantities describe the resource profile of observed successes. They do not estimate causal efficiency because conditioning on success selects different tasks and trajectories across arms. Unconditional costs remain the primary resource comparison.

## D Provenance

The provenance manifest records byte counts and SHA-256 hashes for the frozen study specification, raw and adjudicated datasets and reports, adjudication ledger, post-hoc metrics, masked coding packet, both original coder label files, agreement report, and manuscript: `provenance-manifest.json`.

The adjudication ledger separately hashes each changed row and its supporting evidence. Both original taxonomy-label files were frozen before adjudication.

## E Cross-model coding agreement

The table reports exact agreement, Cohen’s kappa, and Gwet’s AC1 for the two outcome-blind model-coding passes. Kappa is undefined when both coders use one category for every item because the marginals have zero variance. AC1 is a supplemental prevalence-robust coefficient computed with the full pre-specified category vocabulary. Its value of 1.000 for invariant fields is mechanical and supplies no evidence beyond 84/84 exact agreement.

No coefficient interval is reported because these are two fixed model-coding passes used as a prompt and model-sensitivity check rather than sampled human raters. Interpretive fields remain exploratory.

**Table 7:** Timeout and adjudication taxonomy.

| Category                      | Count | Treatment                        | Interpretation                                                                               |
|-------------------------------|-------|----------------------------------|----------------------------------------------------------------------------------------------|
| Official SWE task timeout     | 6     | Scored zero                      | Official evaluation exceeded the frozen 3,600-second task limit.                             |
| Terminal agent-budget timeout | 6     | Scored zero                      | Agent exhausted 1,800 seconds and the completed verifier returned zero.                      |
| ActiveGraph snapshot regrade  | 1     | Hash-identical grader-only retry | Original grader contradicted a receipt-bound parsable submission; no model action was rerun. |

**Table 8:** Cross-model coding agreement. Undefined kappa indicates zero marginal variance.

| Field                           | Exact agreement | Cohen’s kappa | Gwet’s AC1 |
|---------------------------------|-----------------|---------------|------------|
| Primary update target           | 76/84 (90.5%)   | 0.000         | 0.904      |
| Secondary update target         | 47/84 (56.0%)   | -0.081        | 0.543      |
| Representational form           | 84/84 (100%)    | undefined     | 1.000      |
| Transfer scope                  | 63/84 (75.0%)   | 0.470         | 0.717      |
| Abstraction level               | 59/84 (70.2%)   | 0.529         | 0.647      |
| Evidence grounding              | 59/84 (70.2%)   | 0.428         | 0.669      |
| Failure specificity             | 83/84 (98.8%)   | 0.661         | 0.988      |
| Validation strategy             | 82/84 (97.6%)   | 0.592         | 0.976      |
| Consolidation operation         | 84/84 (100%)    | undefined     | 1.000      |
| Executable status               | 84/84 (100%)    | undefined     | 1.000      |
| Anticipated activation          | 84/84 (100%)    | undefined     | 1.000      |
| Counterfactual actionability    | 84/84 (100%)    | undefined     | 1.000      |
| Novelty relative to prior state | 84/84 (100%)    | undefined     | 1.000      |
| Confidence                      | 63/84 (75.0%)   | 0.393         | 0.687      |